400G Ethernet Exploration — An FPGA-Accelerated Cloud Platform for BER Analysis

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Background & Motivation

- **The Demand:** Global infrastructure is migrating to 400G Ethernet to meet surging data traffic.
- **The Technical Gap:** Higher speeds cause severe signal distortions, necessitating new architectures.
- **The Bottleneck:** Software-based simulation is too slow; at a 10^{-11} BER level, it would take 500 years to observe a single error [1].
- **Related Work:** Existing FPGA-accelerated simulator has targeted 200G PAM-4 Ethernet [1].
- **Our Solution:** An AWS FPGA Platform – providing a scalable, portable, and zero-upfront cost environment for accelerated 400G validation.

Project Goal

Our project goal is to build a **fast** and **scalable** channel simulator platform on **AWS** for evaluation of possible **400G PAM-6** Ethernet architectures.

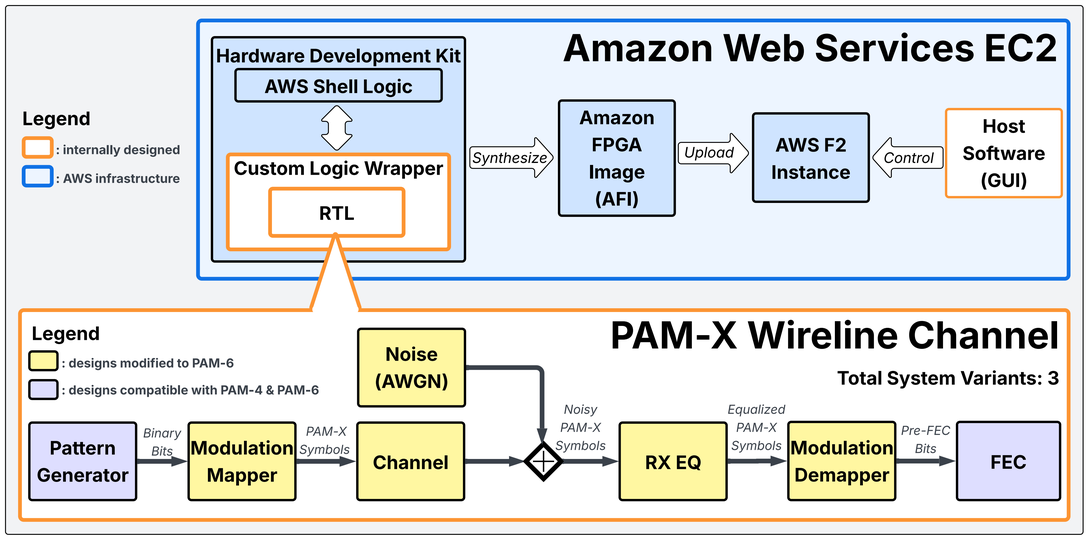
Key Requirements

#	Requirements	Results
1	Supports BER analysis of **PAM-6** modulation under **configurable** noise conditions.	PASS
2	Provides **modular** support for FEC/Receiver architecture(s).	PASS
3	The system must have **faster single-core speed** than the Opal Kelly XEM7310 board.	PASS
4	Enables **software baseline model** that verifies the **correctness** of the RTL model.	PASS
5	Supports baseline PAM4 modulation for **correctness** against reference design.	PASS
6	Supports **parallel-core** execution to enable **scalability**.	PASS

Terminologies

- **BER:** Bit-Error-Rate = $\frac{\text{\# of bit errors}}{\text{\# of total bits transmitted}}$
- **SNR:** Signal-to-Noise Ratio = $\left(\frac{\text{Signal strength}}{\text{Noise level}}\right)_{dB}$
- **FEC:** Forward Error Correction; helps recover the error bits without retransmission through adding additional redundant bits as parity bits
- **PAM-X:** Pulse-Amplitude Modulation; encoding information by varying the voltage into X levels in a channel, thus transmitting more than 1 bit at a time
- **Vivado:** A software tool to synthesize hardware code (RTL) into synthesized design
- **RTL:** Register-Transfer-Level; the "source code" of a digital circuit

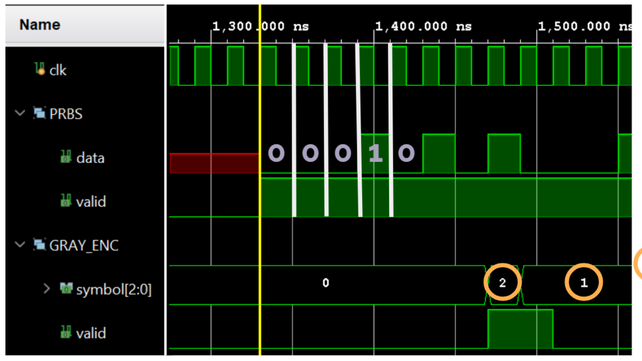
System-Level Diagram



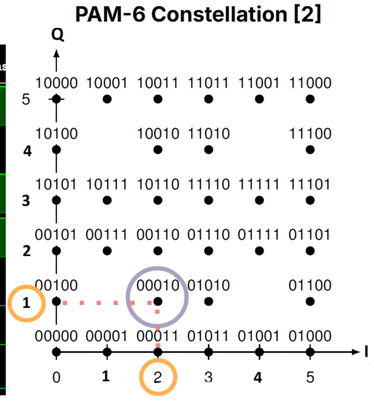
- **Variant 1:** analog AWGN channel and hard-decision Hamming(128,120) inner FEC code.
- **Variant 2:** analog LPF coloured noise channel with convolutional interleaver and hard-decision Hamming(128,120) inner FEC.
- **Variant 3:** 1 + αD channel with AWGN and MLSE equalizer.

Design Validation

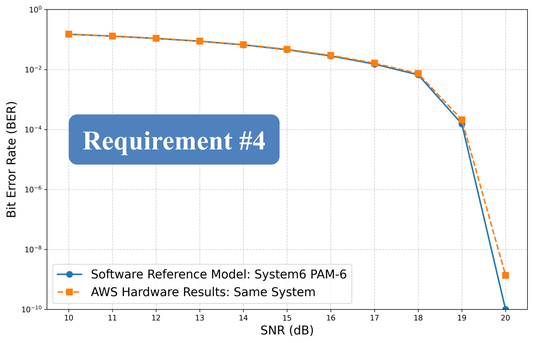
RTL Waveform: Pattern Generator and Modulation Mapper



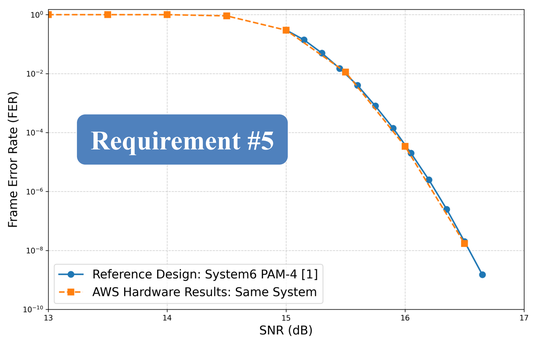
PAM-6 Constellation [2]



RTL Validation: Against Software Reference Model

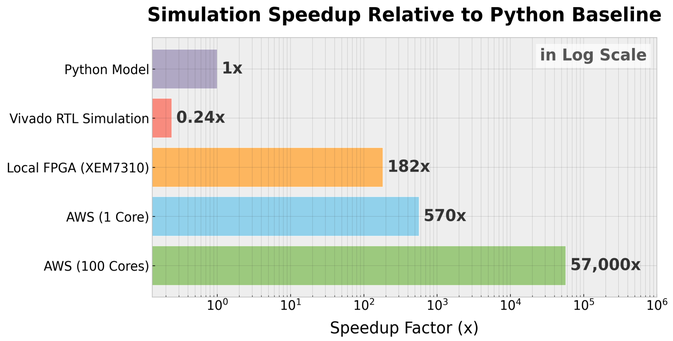


Platform Validation: Against Reference Design [1]



Results

Platform	Throughput	Speedup
Python Model	0.219 Mb/s	1x (baseline)
Vivado RTL Simulation	0.052 Mb/s	0.24x
Local FPGA Board (Opal Kelly XEM7310)	40.0 Mb/s	182x
AWS: 1 Core	**125.0 Mb/s**	**570x**
AWS: 100 Cores	**12500.0 Mb/s**	**57, 000x**



Challenges & Limitations

- No access to **physical FPGA board.**
  → Inserted debug modules to locate the issue.
- **Limited time** to design new blocks (e.g. 2D FEC Code)
  → Focused on expanding reference design.
- **Trade-off:** longer synthesis time is needed to gain more runtime speedup.

Conclusion

- **570x single-core speedup** compared to traditional software simulation; even more with parallelism.
- **Scalable**, **portable**, and **cost-efficient** design.
- Validated system **correctness** via multi-stage testing.
- **User-friendly** interface and documentation.

Future Work

- More **runtime configurability** (modulation switch, receiver block switch) to reduce compile time.
- Bring up more **architectures**, e.g., PAM-8, 2D Reed-Solomon FEC [3].

References

[1] R. Barrie, Analysis of Wireline FEC Performance Using an FPGA Platform, M.A.Sc. thesis, Dept. Elect. & Comput. Eng., Univ. of Toronto, 2024.

[2] G. Böcherer, "On the Performance of PAM-6 Modulation Schemes for High-Speed Ethernet," arXiv preprint arXiv:2205.05460, May 2022. [Online]. Available: https://arxiv.org/pdf/2205.05460

[3] C. Xu, "400 Gbps+ per Lane Transmission with PAM-4, 6, 8," IEEE 802.3 E4AI Ad Hoc Meeting, Jun. 24, 2025. [Online]. Available: https://www.ieee802.org/3/ad_hoc/E4AI/public/25_0624/xu_e4ai_01a_250624.pdf